FARAJ SHEKH

Highway Design Engineer | PhD Scholar @ IIT Gandhinagar

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PROFILE OVERVIEW



Highway Design Engineer and a part-time doctoral research scholar at IIT Gandhinagar with industry experience in executing highway and corridor design projects, including Expressways in Gujarat. Strong expertise in AutoCAD Civil3D for horizontal alignment, vertical profile, plan and profile, corridor modelling, and earthwork calculations; also have knowledge of Sub-Assembly Composer. Experienced in pavement design, including flexible, rigid, semi-rigid, and geosynthetic pavements; traffic engineering surveys; preparation of Feasibility and Detailed Project Reports; and technical schedules for various road projects following IRC guidelines. Also well-versed in end-to-end client and internal stakeholder coordination.

CORE COMPETENCIES & EXPERTISE

- **Highway & Expressway Planning and Design:** Greenfield Expressway Planning, Feasibility Studies, Alignment design and Evaluation, Highway Geometric Design, Corridor Planning, Interchange Design, Junction Improvement, Road Widening and Capacity Augmentation;
- **Pavement Engineering & Design:** Flexible, Semi-Rigid, Rigid and Geosynthetic Pavement Design in accordance with IRC & AASHTO Standards, Pavement Evaluation, Subgrade and Pavement Material Characterisation, FWD and NSV;
- **Competency in Autodesk Civil 3D:** Horizontal Alignment Design, Vertical Profiles, Cross-Sections, Service Road Design, Grade Separators, Interchanges, Ramp Geometry, Cut-and-Fill Assessment, Corridor & Terrain Modelling and Engineering Drawings using Autodesk Civil 3D, Assemblies using Subassembly Composer;
- **Detailed Project Reports & Technical Documentation:** Preparation of Feasibility Reports and DPRs, Inception Reports, Alignment Options Reports, Engineering Drawings, Technical Schedules, RFP Documentation, Contract Schedules (Schedule A, B & C);
- **Traffic Engineering & Transportation Analytics:** Traffic Surveys (CVC, OD, Axle Load, Speed–Delay, Toll Traffic and Turning Movement), Traffic Data Processing and Analysis, Traffic Pattern Assessment, Traffic Forecasting Support, Intersection Traffic Management, Traffic Simulation using SUMO;
- **Highway Asset Assessment & Infrastructure Surveys:** Pavement Condition Assessment, Road Inventory, Bridge and Culvert Inventory and Condition Assessment, Road Geometry Assessment, Reconnaissance Surveys;
- **Coordination with Government Officials & Stakeholders:** Technical Assistance to Roads & Buildings Department (Government of Gujarat), Gujarat State Road Development Corporation (GSRDC), Coordination with Field Divisions and Departmental Officials, Technical Reporting, Data Consolidation and Support for Senior-Level Project Reviews;
- **Engineering Software & Digital Applications:** Autodesk Civil 3D & Subassembly Composer, AutoCAD, IITPAVE, SUMO, Microsoft Excel, Google Sheets, Google Earth, Web-based Digital Monitoring Systems and Reporting Workflows.

Education:

Name of Educational Institutions	Year of Passing	Degree/Exams
Indian Institute of Technology (IIT), Gandhinagar	Aug 2026 - Present	Doctor of Philosophy, Road Asset Management System
L. D. College of Engineering, Ahmedabad	2022-2024	Master of Engineering – Transportation Engineering (9.29 CGPA)
Graduate Aptitude Test in Engineering (GATE)	2022	Qualified (364 score)
L. J. Institute of Engineering and Technology, Ahmedabad	2018-2022	Bachelor of Engineering – Civil Engineering (8.71 CGPA)

Publications:

- "**Enhancing Pavement Durability Through Soil Subgrade Stabilization: A Comparative Cost Analysis**" by Faraj Shekh and Prof. (Dr) Rena N Shukla in International Conference on Sustainable Materials, Environment and Technologies under climate change scenario (SMET-2024)-Springer Nature, 2024.
- "**Blackspot Identification and Hazards on Urban Road: A Case Study of SG Highway, Ahmedabad**" by Faraj Shekh and Prof. (Dr) Rena N Shukla in International Conference on Transportation System Engineering and Management (CTSEM-2023)-Springer Nature, 2023.

Achievements:

- Awarded **"Star Presenter"** by LEA Associates South Asia Pvt. Ltd. in Oct 2026 in Young's Professional Presentation Competition.
- Awarded **"Best Newcomer"** by LEA Associates South Asia Pvt. Ltd. in Aug 2025, along with a half-yearly award.
- Runner-up in academic excellence Gold Medal from Gujarat Technological University in Master of Engineering - Transportation Engineering, 2024

Training & Seminars:

- Pavement Materials and Design as a Course for PhD at IIT Gandhinagar.
- 6-month internship at LEA Associates South Asia Pvt. Ltd. (22nd Apr 2024 to 30th Oct 2024)
- Three-day practical training session focused on flexible pavement design, including hands-on experience in material testing and conducting axle load surveys, Darshan University, Rajkot (6th Nov, 2023 to 8th Nov, 2023)
- Training Program on FWD and NSV Operations and Data Collection and Analysis Process by Infinite Civil Solutions Pvt. Ltd.
- 3-month on-site internship program at Chavda Infra Pvt. Ltd. (03rd Jan 2022 – 03rd Apr 2022)
- 2-month internship program held by L J Institute of Engineering and Technology. (21st May 2021 – 21st July 2021)

Academic Projects

- "Analysis of Flexible Pavement Design alternatives based on Pavement life and Project cost" - The final year Master's dissertation project with the aim of evaluating various alternatives for flexible pavement design with prime consideration of pavement life and project cost.
- "Effect of Glass fiber on rheological properties of VG30 grade bitumen" - The project involved studying the effect of adding glass fibre to VG30-grade bitumen mixes. The prepared mixes were tested using the Marshall Stability test to evaluate their performance.

Software Skills

- Autodesk Civil 3D and Subassembly Composer;
- Autodesk AutoCAD;
- Google Earth;
- QGIS (Basic);
- SUMO (Basic);
- Google Productivity Tools;
- MS Office, MS Excel;
- Pavement Analysis and Design Software: IITPAVE, KGPBACK.

Languages Skills:

Language	Speaking	Reading	Writing
English	Good	Excellent	Excellent
Gujarati	Excellent	Excellent	Excellent
Hindi	Excellent	Excellent	Excellent

Employment Record:

Period of Employment	Duration	Employing Organisation and Title/ Position
Nov 2024 to Present	2 Years	**Employer:** LEA Associates South Asia Pvt. Ltd., New Delhi **Positions Held:** Deputy Engineer (Traffic and Transportation)
22nd Apr 2024 – 30th October 2024	6 Months	**Employer:** LEA Associates South Asia Pvt. Ltd., New Delhi **Position Held:** Intern
Jan 2022- Apr 2022	3 Months	**Employer:** Chavda Infra Pvt. Ltd. **Position Held:** Intern

Summary of the experience: As a Highway Engineer, worked on various important projects involving planning, geometric design, pavement design, traffic engineering & assessment, and preparation of Detailed Project Reports (DPRs) and Feasibility reports. Contributed to the feasibility studies and alignment development of proposed greenfield expressways in Gujarat, including the preparation of alignment options and reports, end-to-end highway geometric design, and identification of existing features like roads, railways, and drainage crossings, windmills, and high-tension lines. Also worked on the Amaravati trunk road connectivity to NH project in Amaravati, Andhra Pradesh, reviewing LiDAR survey data, Greenfield Alignment design inputs, Design of Vertical profiles, cross-sections, corridor modelling, full-fledged trumpet interchange design, and cut-and-fill assessments using Autodesk Civil 3D.

Involved in the preparation of DPRs for road widening and upgradation projects, including the Ahmedabad-Mehsana (8-lane) and Shantipura-Khoraj GIDC (6-lane) corridors. Responsibilities included pavement design, geometric design inputs, preparation of engineering drawings, traffic signage plans, encroachment plans, technical schedules, and contract documentation. Also carried out and analysed various traffic surveys, including Origin-Destination, Axle Load, Speed & Delay, and toll traffic studies, along with road condition surveys.

Provided technical support to the Roads and Buildings Department, Government of Gujarat, through project monitoring, data management and preparation of departmental reports. Monitored state highway development works, damaged bridges, the Core Road Network and the Tourist Connecting Roads Scheme. Developed monitoring formats, a web-based monitoring system, and analytical summaries to support departmental reviews and decision-making. Additionally, contributed to road safety assessments, traffic simulation using SUMO, junction improvement planning, and infrastructure condition assessment works.

Details of Projects and Professional Experience in Highway / Traffic & Transportation etc. or in related field:

1. **Project:** Consultancy Services for Feasibility Study and Preparation of Detailed Project Report for Development of Expressways for Package 1, Package 3 and Package 6;
   Year: September 2025 - Ongoing;
   Location: Gujarat, India;
   Client: Gujarat State Road Development Corporation Ltd. (GSRDC);
   Main project features: The project involves feasibility analysis and preparation of Detailed Project Reports (DPRs) for the development of new greenfield expressway corridors across Gujarat in six packages, and we have been assigned three of them: Packages 1, 3, and 6. The objective of the assignment is to identify optimal alignments and develop technically viable expressway corridors to enhance high-speed regional connectivity, reduce travel time, and support economic growth across key regions of the state. Total Length: 539 km;
   Positions Held: Highway Engineer;
   Activities performed: Worked as part of the design and planning team responsible for greenfield alignment design and development for expressway corridors in Autodesk Civil 3D and pavement design under multiple project packages. Prepared and evaluated horizontal alignments for proposed expressway corridors, considering terrain conditions, settlement avoidance, and integration with existing transportation networks. Contributed to the preparation of the inception report and was responsible for the preparation of the draft alignment options reports, including compilation and analysis of technical inputs such as geographical features, land use characteristics, environmental constraints, and existing infrastructure crossings/along the corridor. Identified and documented locations of required cross-drainage structures, road crossings, railway crossings, and other engineering structures along the proposed alignments. Assisted in preparing technical datasets and engineering inputs required for feasibility assessment and DPR preparation. Supported the compilation of design information and preparation of technical documentation, maps, and alignment-related analysis.
   Through this assignment, gained experience in greenfield corridor planning, expressway alignment design, feasibility assessment, and technical documentation for such large-scale highway infrastructure development projects.

2. **Project:** Technical Assistance to the Policy and Planning Unit, Roads and Buildings Department, GoG (R&BD);
   Year: May 2025 - Ongoing;
   Location: Gujarat, India;
   Client: Roads and Buildings Department, GoG (R&BD);
   Main Project Features: Strengthening the Policy and Planning Unit (PPU) and supporting the Roads & Buildings Department, GoG, in project planning, monitoring, and reporting mechanisms for major road infrastructure schemes and initiatives across the state. The assignment focuses on improving institutional capacity to monitor ongoing highway and bridge projects, develop standardised reporting systems, prepare departmental reports, and assist with the evaluation and tracking of flagship infrastructure schemes. The project also involves development of structured data systems for real-time monitoring of works undertaken by various field divisions of the department;
   Positions Held: Highway Engineer;
   Activities Performed: Provided technical assistance to the PPU, Roads & Buildings Department in monitoring the progress of infrastructure works across Gujarat. Contributed to the preparation of the Annual Report of the Roads & Buildings Department for FY 2024-25, including compilation,

analysis, and coordination with departmental divisions for data collection. Developed structured data collection formats and monitoring templates, and managed Google Sheets-based tracking systems to streamline reporting and improve coordination between the secretariat and field divisions. Also developed a web-based monitoring system. Supported the monitoring of Garvi Gujarat High-Speed Corridor initiative of 1100 km, including preparation of review formats and consolidation of project progress information from multiple divisions. Played a key role in preparing Superintending Engineer (SE) review presentations, summarising project performance indicators and highlighting key monitoring observations for departmental review meetings. After the Gambhira bridge incident, carried out daily monitoring of 158 and later 173 damaged bridges across the state, compiled status updates received from divisions, and developed automated monitoring abstracts for efficient tracking and reporting to senior officials. Played a key role in monitoring the Tourist Connecting Roads Scheme, a flagship initiative to improve connectivity to major tourism destinations in Gujarat, Core Road Network, White Topping Scheme, and PBMC/STMC scheme.

Through this assignment, gained experience in infrastructure project monitoring, institutional reporting systems, highway program management, and coordination with government officials for large-scale road infrastructure development programs.

3. **Project:** Concept Plan for Traffic Management at Ch-0 Intersection in Gandhinagar on NH-147 and SH-71;
   Year: June 2025 – February 2026;
   Location: Gujarat, India;
   Client: Capital Project Division 3, Roads and Buildings Department, Government of Gujarat;
   Main Features: The assignment involved the preparation of a concept-level traffic management and geometric improvement plan for the CH-0 intersection in Gandhinagar. The assignment focused on evaluating the existing traffic conditions, identifying operational issues, and developing an optimised intersection layout to improve traffic flow, enhance safety, and accommodate future traffic demand;
   Position Held: Highway Engineer;
   Activities Performed: Participated as part of the design team in the preparation of the concept plan for the CH-0 intersection improvement. Responsible for the design of the proposed intersection layout, including preparation of horizontal alignments and vertical profiles in Autodesk Civil 3D. Prepared cross-sections for the proposed intersection design. Assisted in the conceptual geometric design of the intersection, focusing on safe traffic movements, improved turning radius, and better operational efficiency. Responsible for the preparation of design drawings and drafting and compilation of the report. Coordinated with team members for integrating various technical inputs and consolidating them into the final project documentation.

   The assignment provided hands-on experience in grade-separated junction design, traffic flow improvement planning, geometric design development, and preparation of technical reports for highway intersection improvement projects.

4. **Project:** Detailed Project Report Preparatory (DPR) and Procurement Services for Ahmedabad-Mehsana (4L) Toll Road SH-41;
   Year: July 23 - Feb 2026;
   Location: Gujarat, India;
   Client: GRICL and Roads and Buildings Department, GoG;
   Main Project Features: Project Preparatory (DPR) and Procurement Services; Upgradation and strengthening of the Ahmedabad-Mehsana corridor (SH-41) under the DBFOMT model on a modified annuity basis from 4 lanes to 8 lanes with service roads and Grade Separators. The project includes geometric improvements, pavement strengthening, development of service roads, intersection

improvements, forest clearance, and preparation of the detailed project report and detailed engineering inputs required for contract documentation and implementation under the DBFOMT model; Length: 49 km, 8 lanes, Project Cost: 2,617 Crores;
Position held: Highway Engineer**;**
Activities performed: Responsible for the preparation of contract schedules documentation and engineering inputs for the Ahmedabad–Mehsana road upgradation project. Prepared Schedule A, B, and C in accordance with project requirements, defining the scope of works, project assets, and maintenance obligations under the DBFOMT framework. Supported the design review and finalisations of road profiles, including elevated corridors, grade separators, and ramps. Prepared cross-sections and design inputs for the main carriageway and service roads profiles. Extracted and compiled existing ground levels and roadside feature data for service roads and adjoining infrastructure. Assisted in preparing design drawings required for DPR and contract documentation. Through this assignment, contributed to highway design refinement, preparation of contract schedules, geometric design updates, and documentation for implementation of a state highway project under a PPP-based model.

5. **Project:** Consultancy Services for Preparation of Detailed Project Report (DPR) for Connecting Amaravati Trunk Infrastructure Roads - E5, E11, E13 and E15 to NH-16 in Amaravati, Capital City of Andhra Pradesh, India;
   Year: Dec 24 - Mar 2025;
   Location: Amaravati Capital City Region, Andhra Pradesh, India;
   Client: Amaravati Development Corporation Limited (ADCL) and APCRDA-Andhra Pradesh Capital Region Development Authority, Andhra Pradesh;
   Main project features: Development of 4/6-lane greenfield trunk roads connecting Amaravati's proposed E5, E11, E13 and E15 corridors with NH-16, including geometric design, grade-separated interchange planning and Feasibility and DPR preparation, Hill cutting, Limited RoW in Forest, Length:12 km;
   Positions held: Highway Engineer;
   Activities performed: Responsible for the preparation of vertical profiles for the finalised alignment and cross-sections for the proposed greenfield road corridors connecting the trunk infrastructure roads to NH-16 in Autodesk Civil 3D. Carried out surface data analysis and terrain modification, including cleaning of survey data, correction of surface points, and modelling of the existing hill. Extracted rail top levels and relevant data for preparation of General Arrangement Drawings for ROB. Responsible for the geometric design of a trumpet interchange connecting the proposed E13 corridor with NH-16. Supported preparation of design drawings and layout plans for the interchange configuration. Modelled the corridor, calculated cut and fill quantities, assisted in developing engineering drawings, and supported compilation of technical inputs for the DPR report.
   Through this project, gained extensive experience in greenfield highway design, interchange design, terrain data processing, corridor modelling, quantity estimation, and DPR preparation for major urban infrastructure connectivity projects.

6. **Project:** Preparation of Detailed Project Report (DPR) for Shantipura Circle to Khoraj GIDC Junction, section of Ahmedabad-Viramgam-Maliya State Highway;
   Year: July 2024 - December 2024;
   Location: Gujarat, India;
   Client: Gujarat State Road Development Corporation (GSRDC);
   Main project features: Detailed Project Report for Upgrading of Shantipura to Khoraj State Highway-17 a Section of Ahmedabad-Maliya Toll Road, Length: 28.87 km, 6 lanes;

Positions held: Intern / Highway Engineer;
Activities performed: Performed and managed traffic surveys and site reconnaissance surveys to understand existing traffic characteristics, roadway conditions, and surrounding development along the project corridor. Assisted in compiling and analysing survey data to support traffic assessment and design considerations for the DPR. Worked as part of the design and drafting team, preparing road signage plans, encroachment plans, and cross-section drawings for the project corridor. Responsible for the preparation and compilation of DPR documentation, including drafting of report sections, integration of design drawings, and coordination with other technical teams involved in the project. Prepared technical schedules and RFP for the Project and assisted in the concessionaire agreement preparation and review.
Through this assignment, gained practical experience in traffic data collection, highway widening design support, preparation of engineering drawings, and documentation of DPR reports for state highway capacity augmentation projects.

7. **Project:** Consultancy Services for "Formulation of High Priority - Long Term Road Development Program";
Year: February 2024 - January 2025;
Location: Gujarat, India;
Client: Roads and Buildings Department, GoG;
Main project features: Formulate a theme-based long-term road development program for the state. Support R&BD in the appraisal of project/s to the Government of India (GoI) for multilateral funding. Preparing phased financing and roll-out plan for programme implementation; Length – 17,668 km, 6-Lane- 1,078 km, 4-Lane-4,670 km, 2L-10,455 km, IL-726 km, SL-740 km;
Positions Held: Intern;
Activities performed: Participated in the High-Priority Long-Term Road Development Program (HPLTRDP), involving condition assessment of approximately 17,668 km of road network across Gujarat. Conducted field investigations in South Gujarat and parts of Saurashtra, covering pavement conditions, road geometry, bridges, culverts, roadside features and speed–delay studies. Also contributed to the processing and analysis of Origin-Destination (OD) surveys at around 150 locations and Axle Load Surveys at approximately 75 locations, along with compilation of survey data and preparation of supporting information for the final project report.

Certification:
I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience. I understand that any wilful misstatement described herein may lead to my disqualification or dismissal if engaged.



Location: Gandhinagar Signature
Date: 04 Nov 2026